

Lecture 4: Turing machines

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Lecture 4

Recall: **Pumping Lemma**

(and how to prove that a language is not regular)

Pumping Lemma and its Proof Sketch

If A is a regular language, then there is a number p (the pumping length) such that, for every string s in A of length at least p , there exists a division of s into three pieces, $s = xyz$ s.t.

1 for each $i \geq 0$, $xy^iz \in A$

2 $|y| \geq 1$, and

3 $|xy| \leq p$.

Pumping Lemma and its Proof Sketch

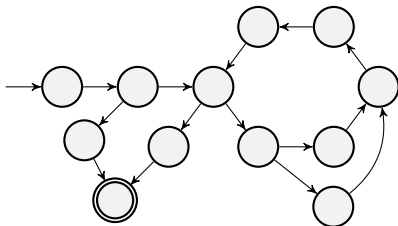
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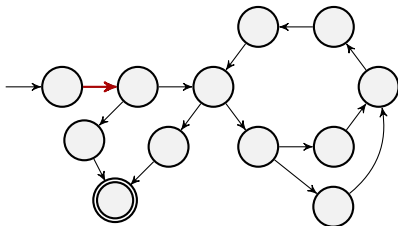
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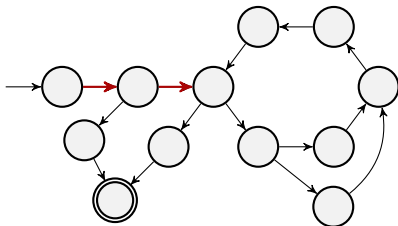
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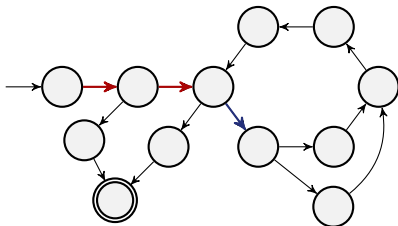
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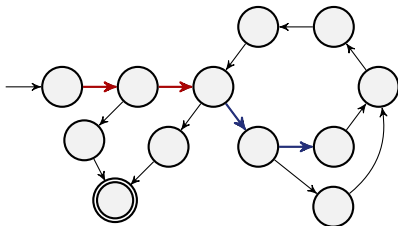
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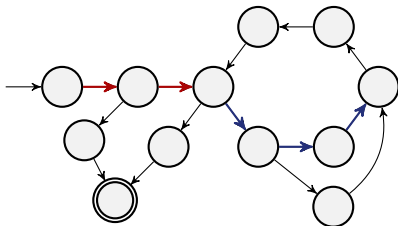
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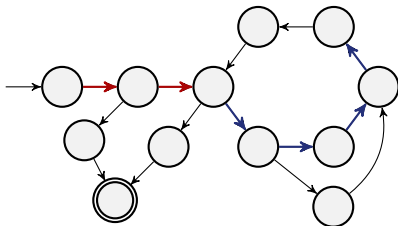
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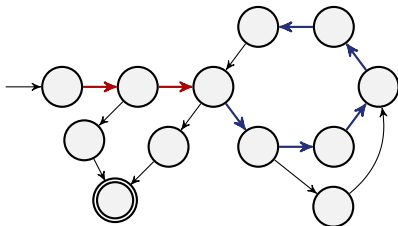
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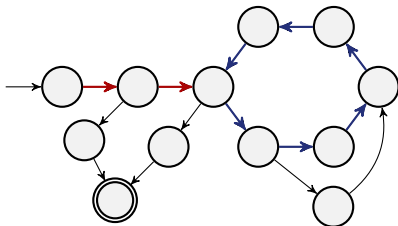
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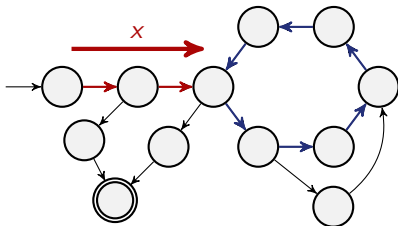
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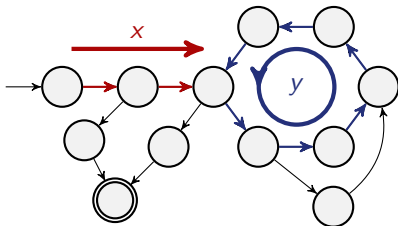
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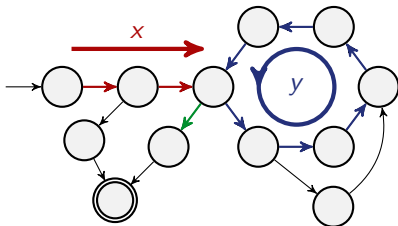
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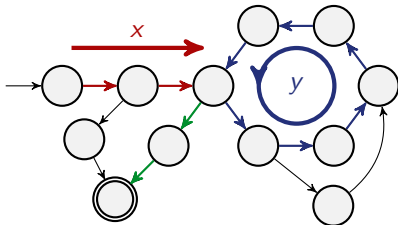
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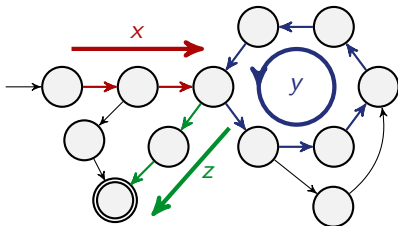


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F is not regular!

Proof: (by contradiction)

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All strings don't work! Fun part is guessing which string to pick
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Pumping lemma tells us there is such a decomposition – we can't choose it! Your reasoning should work for any decomposition

- ▶ Since $|xy| \leq p$ and $|y| \geq 1$, $y = 0^k$ for some $k > 0$
- ▶ According to pumping lemma, $xy^2z \in F$
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- ▶ **Contradiction!**

Part I of course

Lec 1: DFA and Regular Languages

Lec 2: NFA and its equivalence to DFA

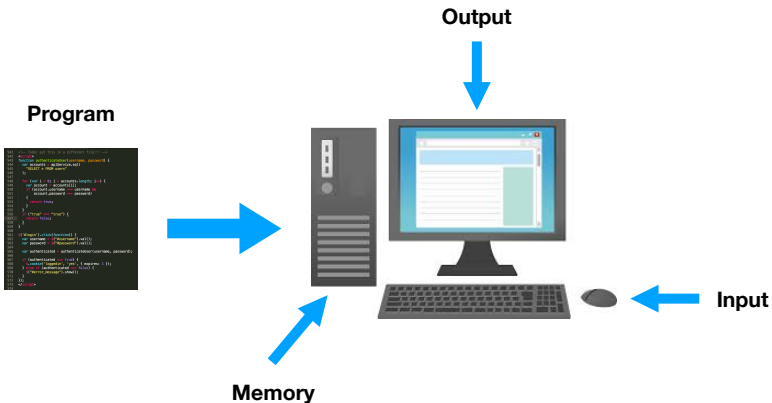
Lec 3: Non-regular languages and the Pumping Lemma

- ▶ $\{w \in \{0,1\}^* : w \text{ has the same number of 0s and 1s}\}$ cannot be recognized by DFAs
- ▶ This seems like a problem with DFAs

What's missing?

What is a computer?

What is a computer?



What is a computer?

finite size (independent of input)

Program

```
1 // Program to calculate the sum of first n natural numbers
2 // Input: n = 5
3 // Output: 15
4
5 #include <iostream>
6 using namespace std;
7
8 int main()
9 {
10     int n;
11     cin >> n;
12     int sum = 0;
13     for (int i = 1; i <= n; i++)
14         sum += i;
15     cout << sum << endl;
16     return 0;
17 }
```



Memory

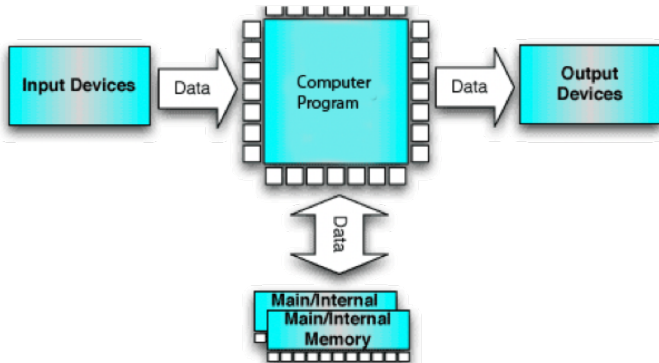
amount depends on input

Output

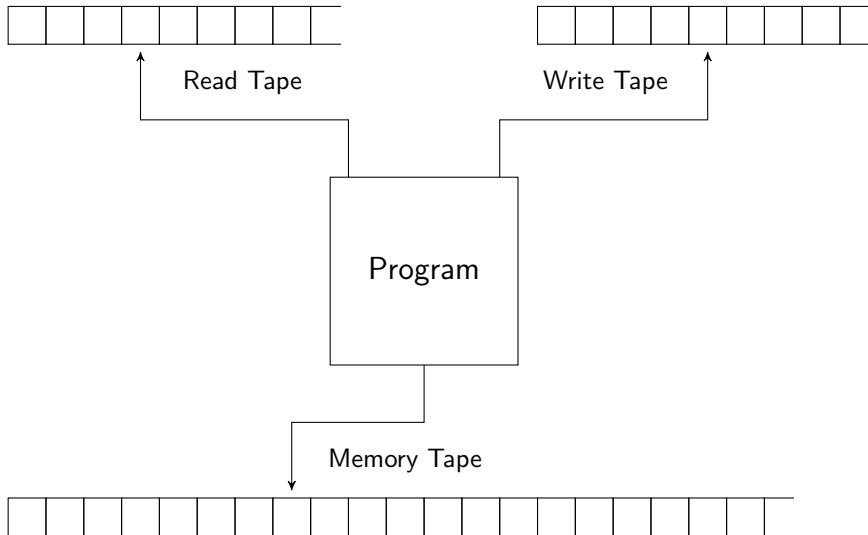


Input

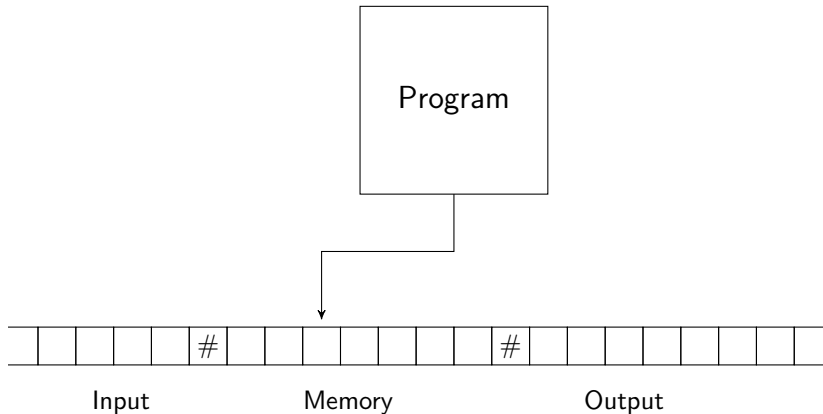
Abstractly



More abstractly



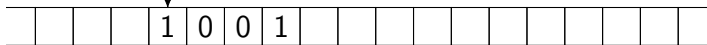
Single tape seems enough...



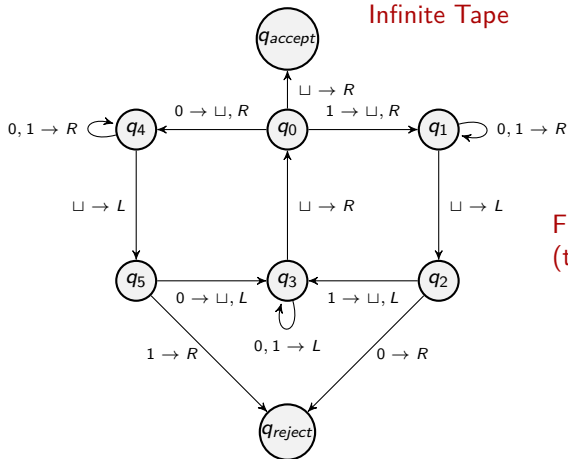
The Turing Machine

Head (with current state)

q_0



Infinite Tape

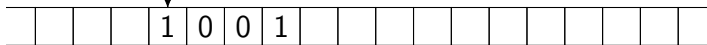


Finite state control
(the algorithm)

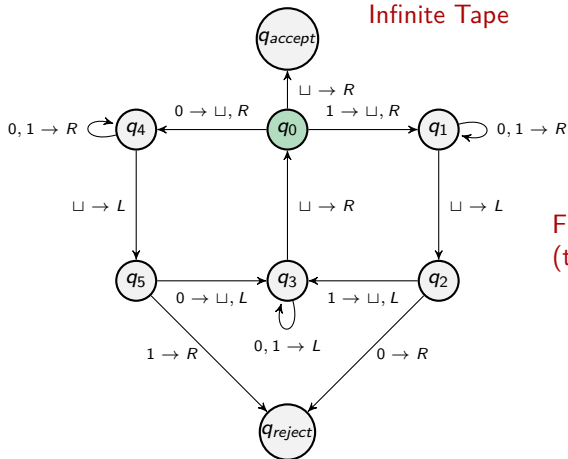
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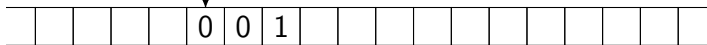


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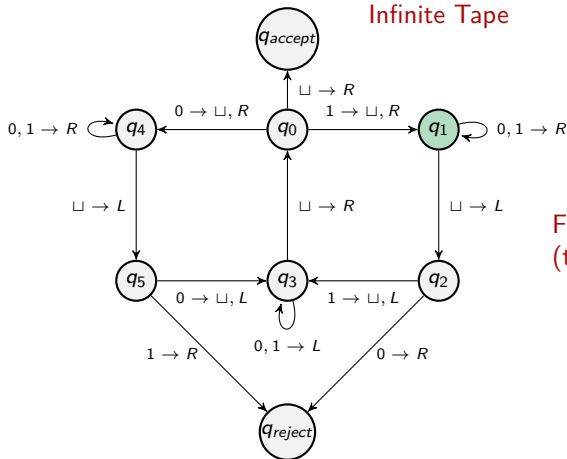
The Turing Machine

Head (with current state)

q_1



Infinite Tape

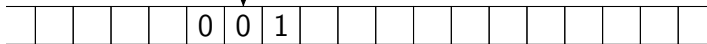


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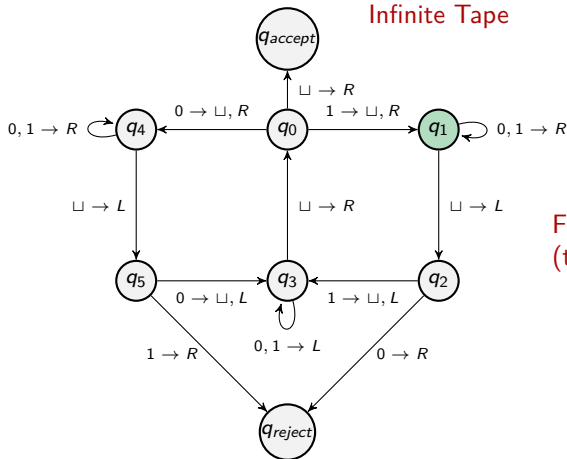
The Turing Machine

Head (with current state)

q_1



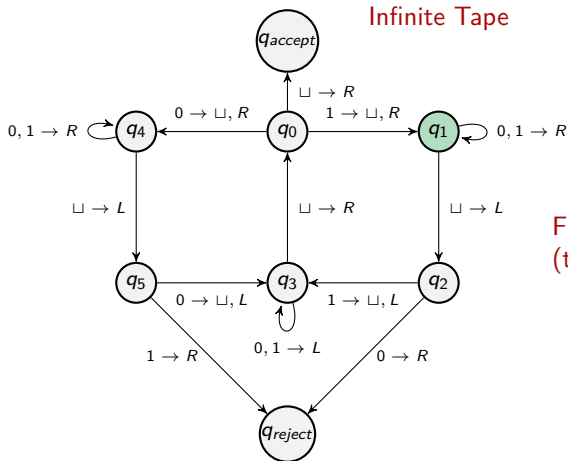
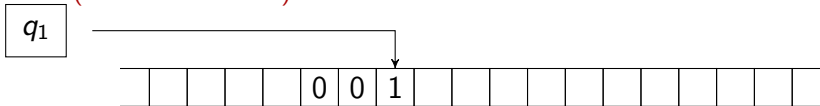
Infinite Tape



Finite state control
(the algorithm)

The Turing Machine

Head (with current state)



Finite state control
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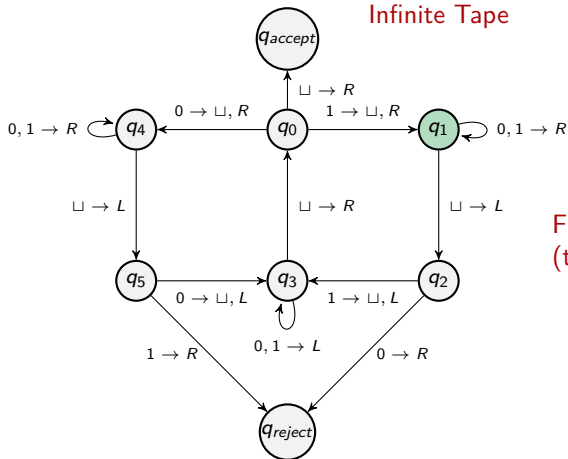
The Turing Machine

Head (with current state)

q_1



Infinite Tape



Finite state control
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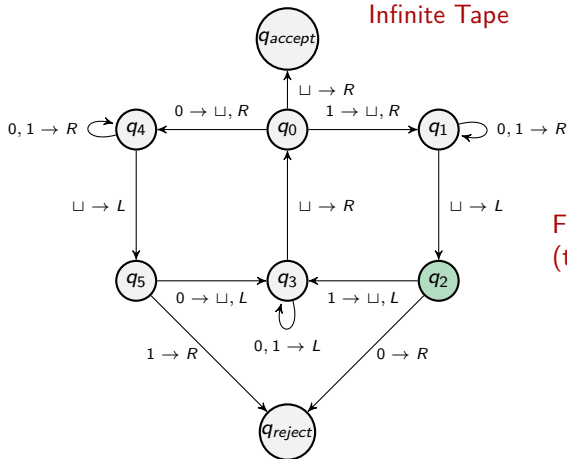
The Turing Machine

Head (with current state)

q_2



Infinite Tape



Finite state control
(the algorithm)

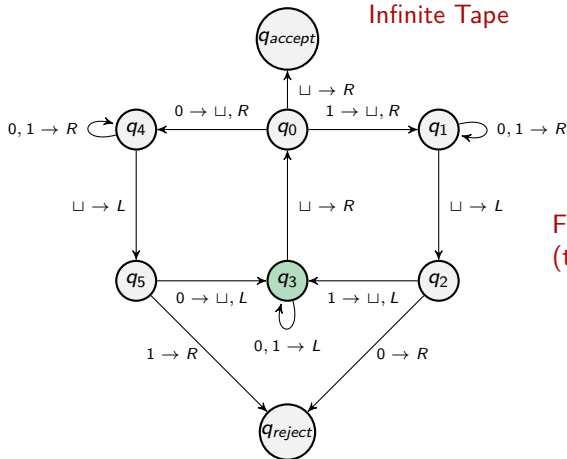
The Turing Machine

Head (with current state)

q_3



Infinite Tape



Finite state control
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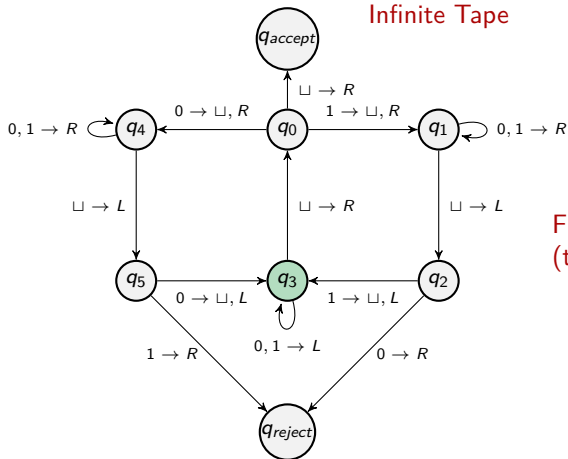
The Turing Machine

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q_3



Infinite Tape

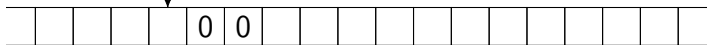


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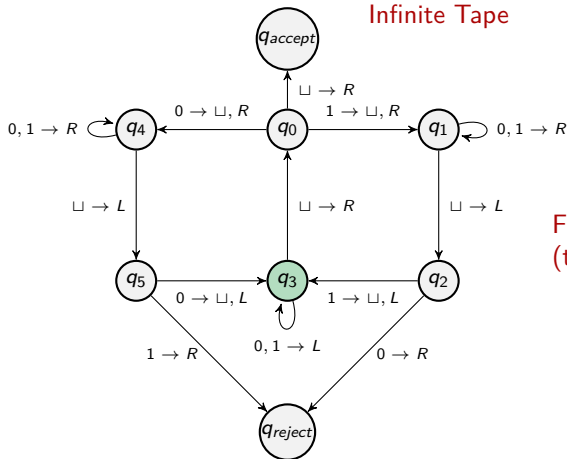
The Turing Machine

Head (with current state)

q_3



Infinite Tape



Finite state control
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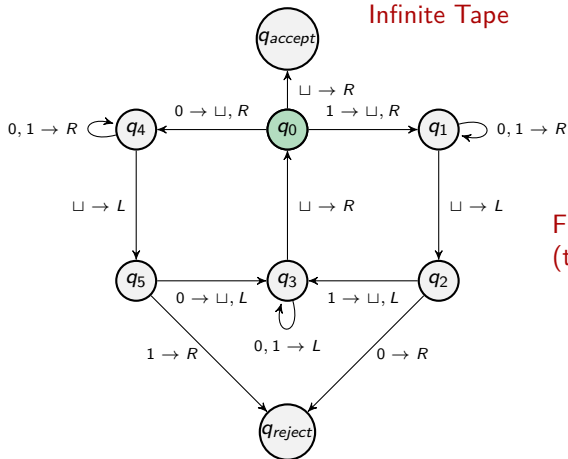
The Turing Machine

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Infinite Tape

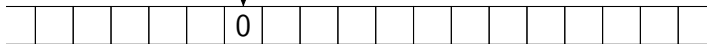


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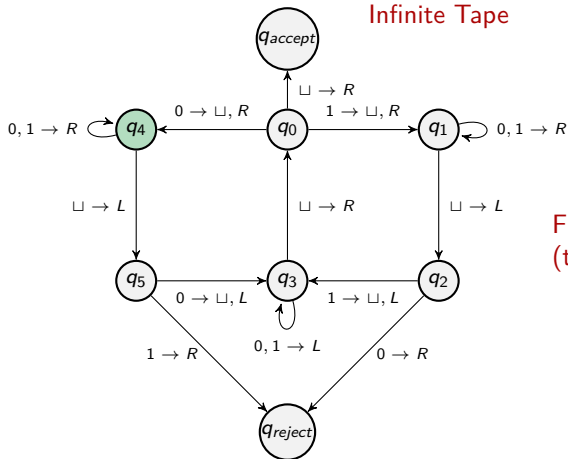
The Turing Machine

Head (with current state)

q_4



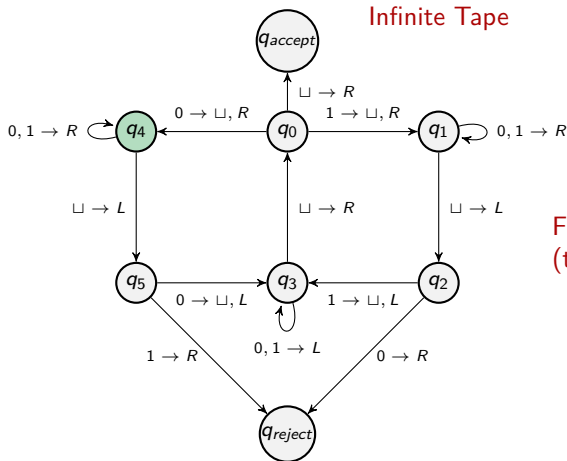
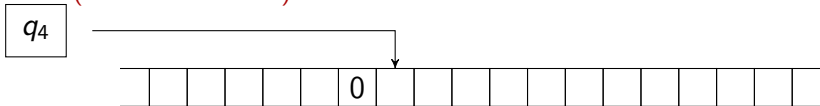
Infinite Tape



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The Turing Machine

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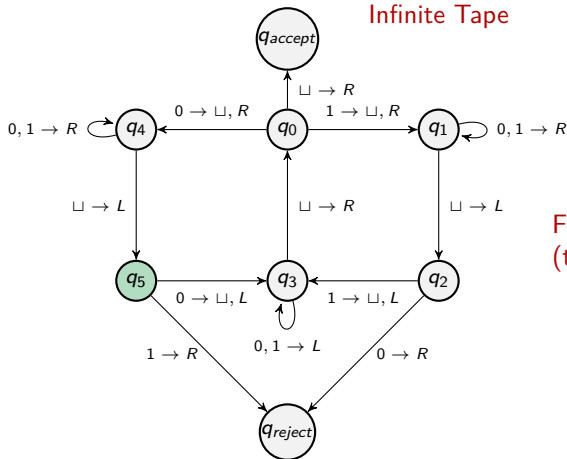
The Turing Machine

Head (with current state)

q_5



Infinite Tape

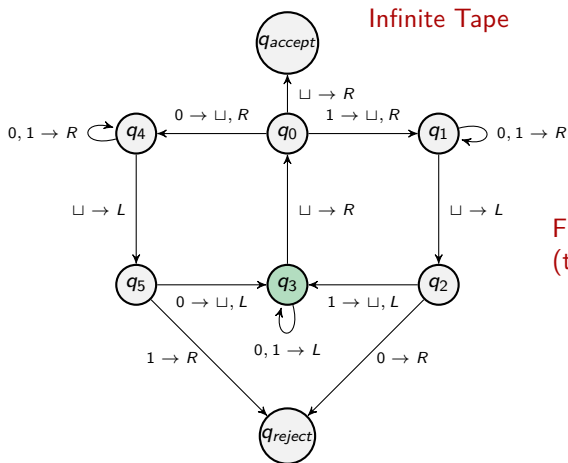


Finite state control
(the algorithm)

The Turing Machine

Head (with current state)

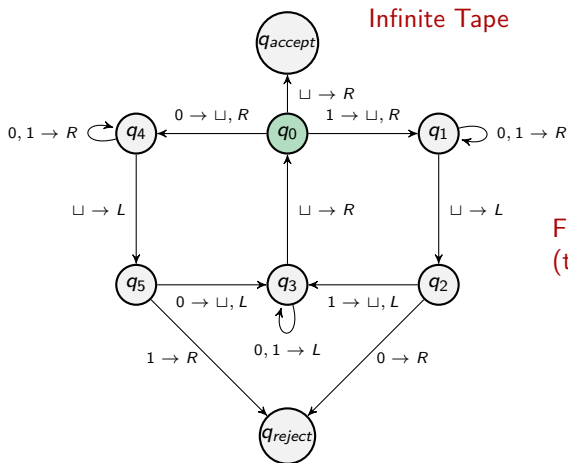
q_3



The Turing Machine

Head (with current state)

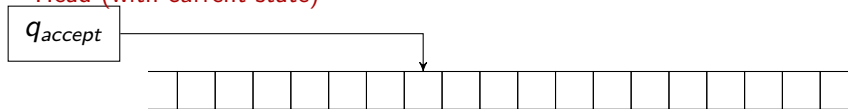
q_0



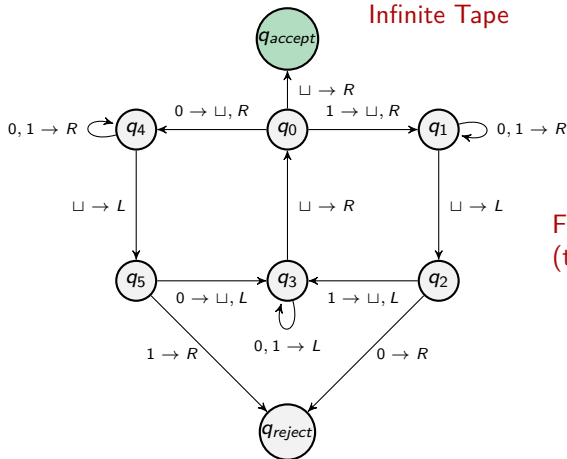
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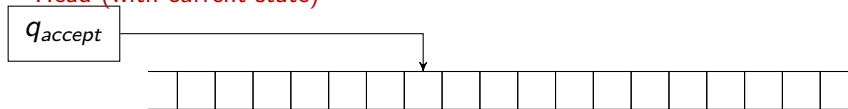
Infinite Tape



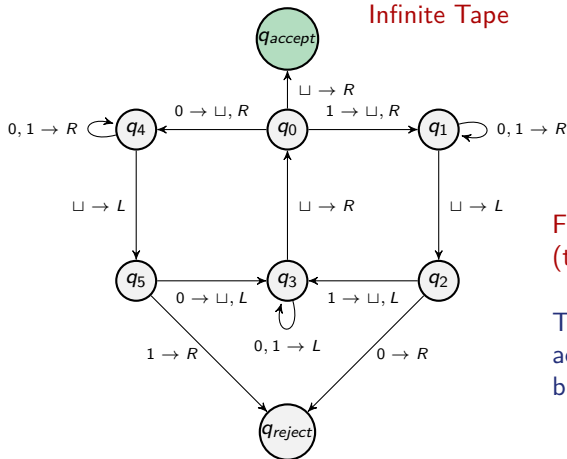
Finite state control
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The Turing Machine

Head (with current state)



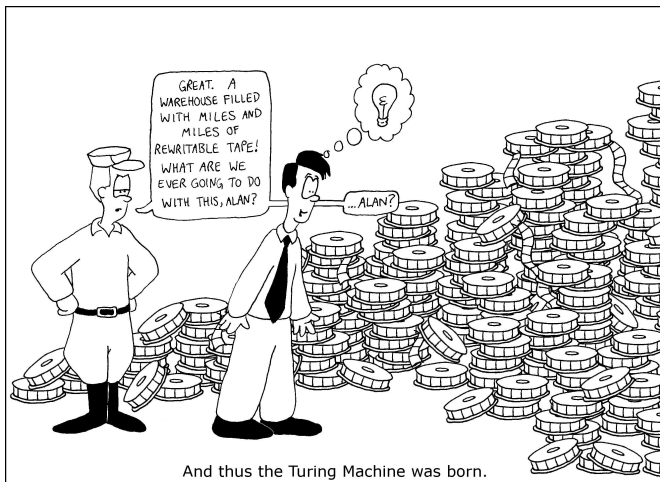
Infinite Tape



Finite state control
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This Turing machine
accepts even length
binary palindromes

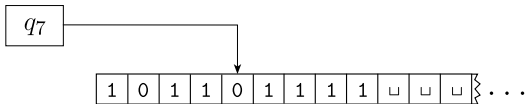
Finite size program, larger and larger instances
⇒ Infinite Tape!



Finite automata vs Turing machines

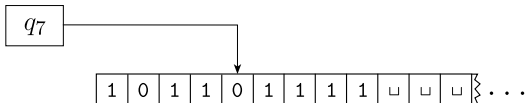
- 1 A Turing machine can both write on the tape and read from it
- 2 The read-write head can move both to the left and to the right
- 3 The tape is infinite
- 4 The special states for rejecting and accepting take effect immediately

Turing Machine



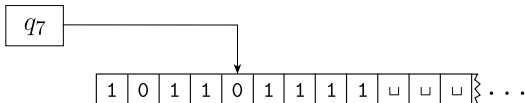
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Turing Machine



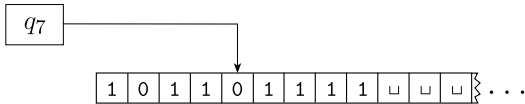
- ▶ Infinite tape
- ▶ Tape alphabet contains input alphabet plus $\square$ (blank symbol) plus maybe more symbols

Turing Machine



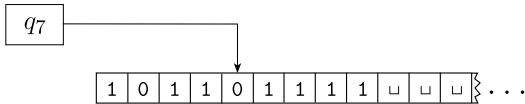
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Turing Machine

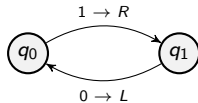
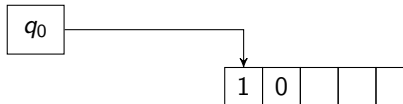


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Turing Machine



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- ▶ Head has states (corresponding to the finite control automata)
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- ▶ Remaining states “*computation in progress*”
- ▶ May never reach an accept state. **May never halt!**



Formal Definition of a TM

A **Turing Machine** is a 7-tuple $(Q, \Sigma, \Gamma, \delta, q_0, q_{accept}, q_{reject})$, where Q, Σ, Γ are all finite sets and

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- 6 $q_{\text{accept}} \in Q$ is the accept state, and
- 7 $q_{\text{reject}} \in Q$ is the reject state, where $q_{\text{accept}} \neq q_{\text{reject}}$.

$\{w : w \text{ has an equal number of 0s and 1s}\}$

Easy (efficient) to write an algorithm to count number of 1s and 0's —
let's try to implement on a TM

(In this Part II of the course, we do not care about running time. . .)

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Our approach: check for each 0 (or 1) there is a corresponding 1 (or 0)

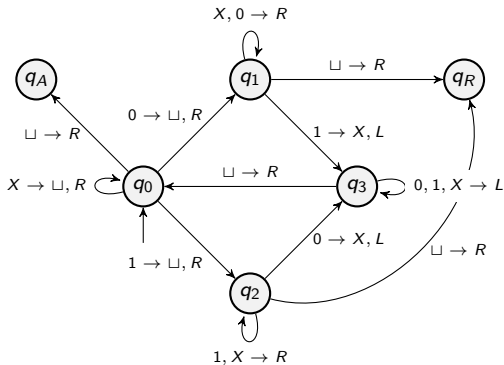
- ▶ Scan the input from left to right
- ▶ Whenever we encounter an uncrossed 0 or 1, we “remove” it and proceed right to find a corresponding 1 or 0 that we cross
- ▶ We keep doing this (2) until either we cross off all the letters (accept) or we fail to find a pair for one of the letters (reject)

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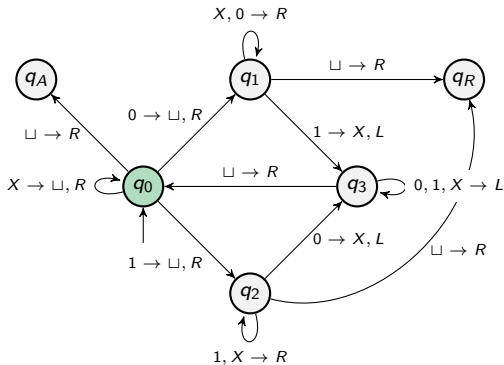


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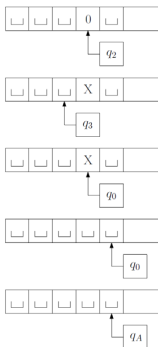
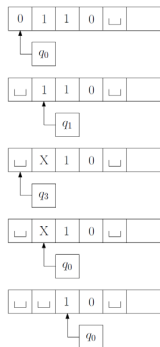
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Execution of the TM on 0110

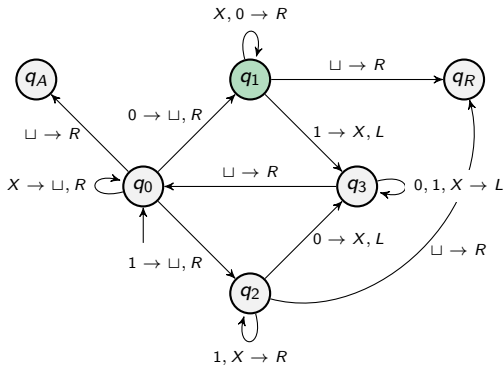


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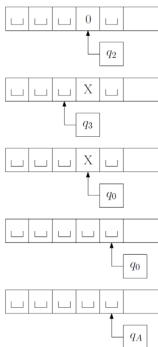
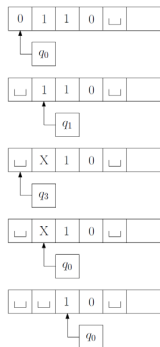
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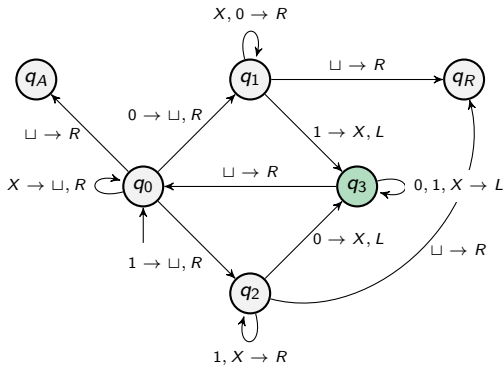


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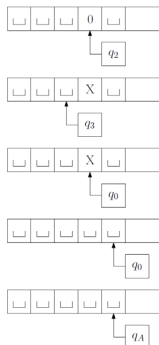
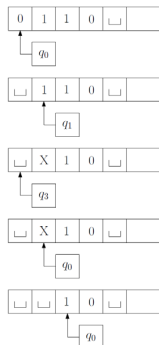
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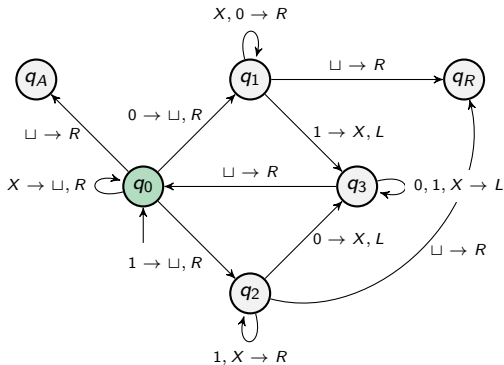


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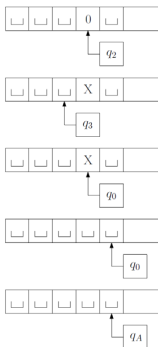
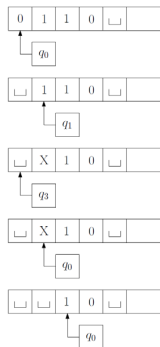
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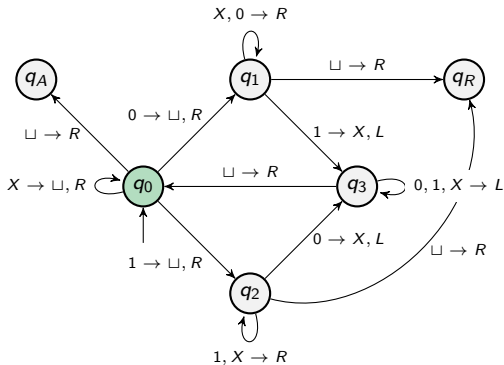


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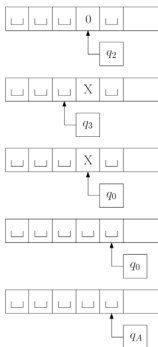
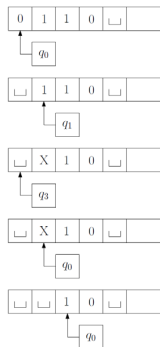
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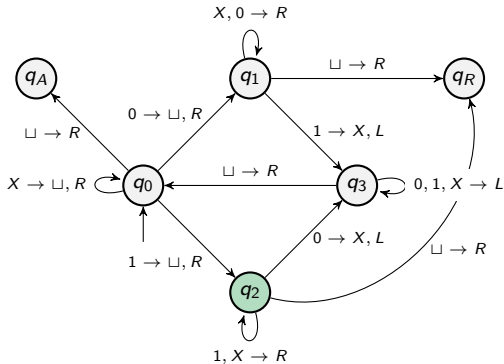


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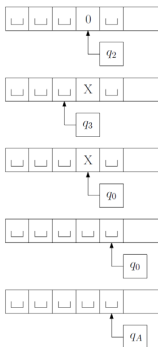
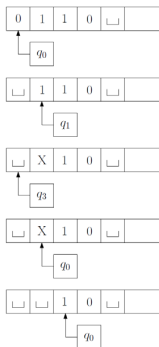
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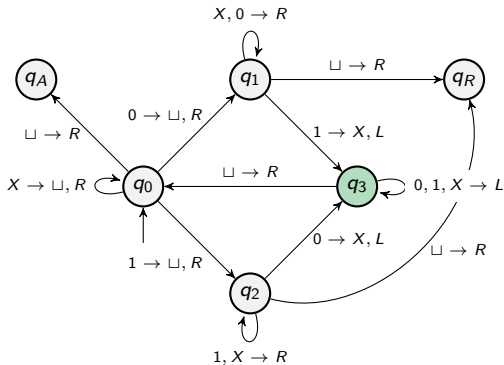


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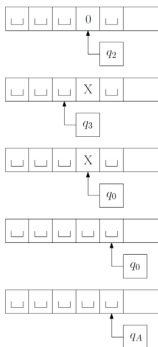
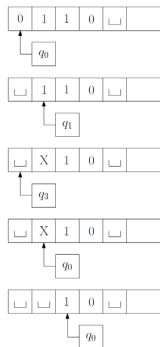
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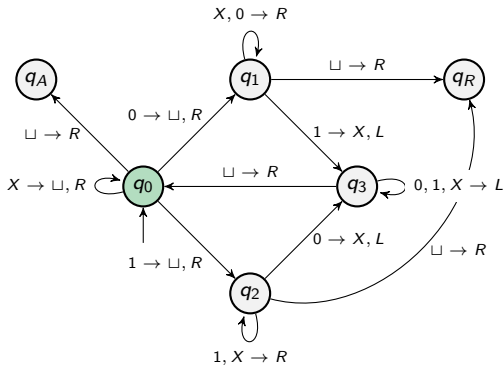


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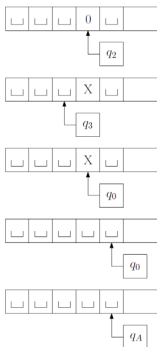
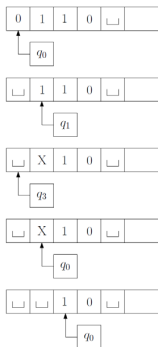
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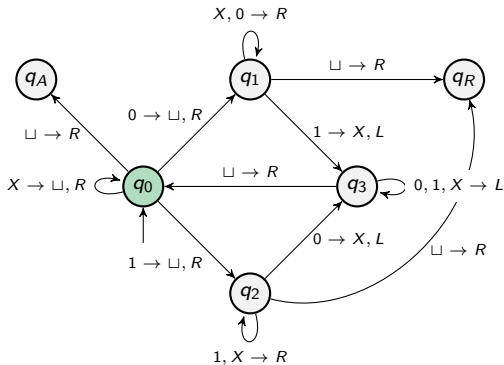


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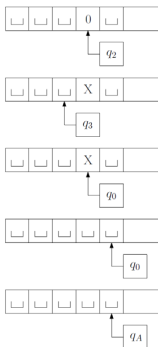
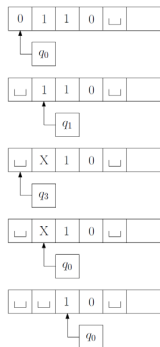
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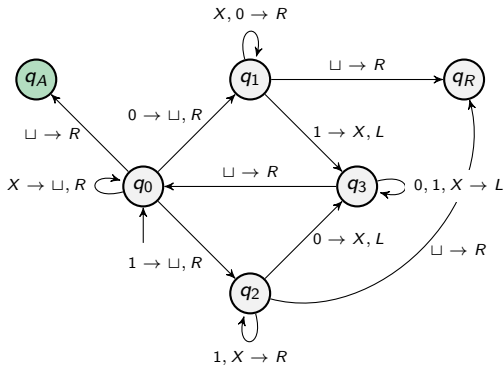


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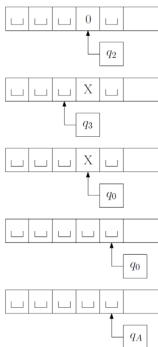
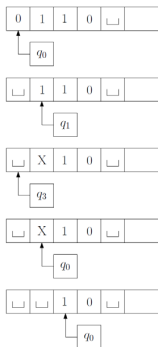
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Representation: We write $uq v$ where $u, v \in \Gamma^*$ and $q \in Q$ for the configuration where

- ▶ current state is q
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(Cells whose contents are unspecified are blank. If $u = \varepsilon$ then Head at leftmost cell.)

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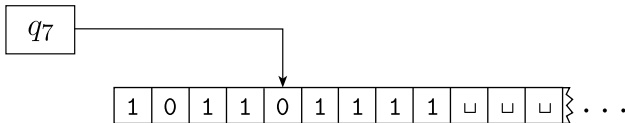
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Example: A TM with configuration $1011q_701111$



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- ▶ Obtain new configurations $C_2, C_3, \dots$ by valid moves/transitions

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- ▶ Starting configuration is $C_1 = q_0 w$ on input $w \in \Sigma^*$
- ▶ Obtain new configurations $C_2, C_3, \dots$ by valid moves/transitions
- ▶ Accept and halt if a configuration with the state q_{accept} is reached

Transitions: Given configuration $uaq_i bv$ where $a, b \in \Gamma$, $u, v \in \Gamma^*$ and state $q_i \in Q$, we move to

- ▶ $uq_j acv$ if $\delta(q_i, b) = (q_j, c, L)$
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What if the computation doesn't halt (i.e., loops)?

Does it mean some configuration is repeated?

Turing-Recognizable/Decidable Languages

A TM machine M **recognizes** a language $L \subseteq \Sigma^*$ iff for all inputs $w \in \Sigma^*$:

- 1 If $w \in L$ then M accepts w and
- 2 If $w \notin L$ then M **does not halt or it rejects** w

Such languages are called **(Turing)-Recognizable**

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A TM machine M **decides** a language $L \subseteq \Sigma^*$ iff for all inputs $w \in \Sigma^*$:

- 1 M halts on w , and
- 2 M accepts w iff $w \in L$

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$$\{0^{2^n} : n \geq 0\}$$

TM that decides this language: On input string w

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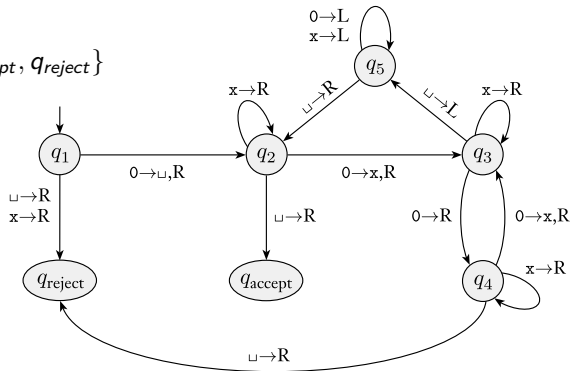
- 1 Remove first 0
- 2 Sweep left to right across the tape, crossing off every other 0
- 3 If in stage 1 the tape contained a single 0, *accept*
- 4 If in stage 1, the tape contained more than a single 0 and the number of 0s that we crossed out was odd, *reject*
- 5 Return the head to the left-hand end of the tape
- 6 Go to stage 1

Each iteration of stage 1 cuts the number of 0s in half.

$$Q = \{q_1, q_2, q_3, q_4, q_5, q_{\text{accept}}, q_{\text{reject}}\}$$

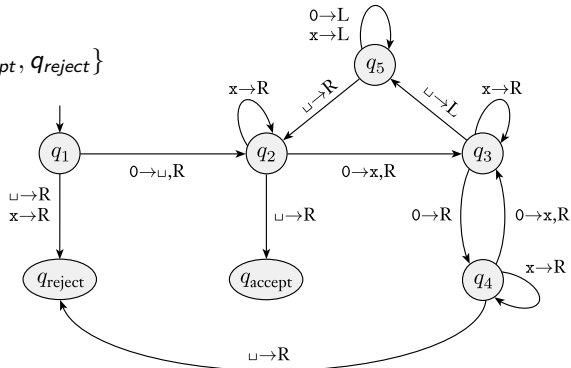
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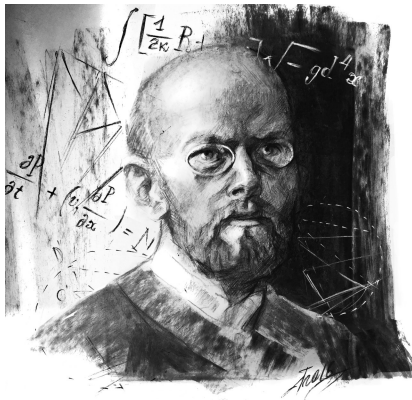
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Example on input 0000:

 $q_1 0000$
 $\sqcup q_2 000$
 $\sqcup x q_3 00$
 $\sqcup x 0 q_4 0$
 $\sqcup x 0 x q_3 \sqcup$
 $\sqcup x 0 q_5 x \sqcup$
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Equivalence to Other Models and History

In 1900, David Hilbert raised 23 mathematical problems



Hilbert's tenth problem

Given a Diophantine equation, **devise a process** according to which it can be determined in a **finite number of operations** whether the equation is soluble in integers.

- ▶ A Diophantine equation is a polynomial equation with integer coefficients and finite number of unknowns.
- ▶ E.g. $3x^2 - 2xy - y^2z - 7 = 0$ has solution $x = 1, y = 2, z = -2$

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Problem: in 1900 there was not a definition of what an algorithm is!

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The precise definition of algorithms allowed Yuri Matijasevic in 1970, building on the work of Martin Davis, Hilary Putnam, and Julia Robinson, to resolve Hilbert’s tenth problem. **No such algorithm can exist!**

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- ▶ And so on

Next week: **Undecidability**